

Midterm Exam — Practice Problems

Reference: Banda-Barragán, W. E. (2026). *Computational Physics 1: UC.1, UC.2 and UC.3 lecture notes*. GitHub.

Course repository: <https://github.com/wbandabarragan/computational-physics-1>

The Midterm Written Exam will have 2 sections:

1. **Multiple-choice and questions:** short concept questions, hand-written pseudocode and workflow diagrams, of the kind listed in Parts A and C below (as in the quizzes).
2. **Code and problems:** questions on the code, problems and analyses of the take-home exam notebook: [exams/midterm-exam.ipynb](#).

How to work through it

- **The written exam is taken without a computer, a notebook or a phone.** You should practice designing algorithms and writing your own code.
- **Practise writing pseudocode by hand:** numbered steps, one action per line, loops and conditionals explicit, every function named with its inputs and its outputs.
- Attempt coding yourself before looking anything up. Submission is not needed.

Part A — What to study

UC.1 — Introduction to computer science

- 101.** Colab and notebooks: cell types, runtime, resources, Linux commands and Python objects, mounting Drive.
- 102.** Linux: the filesystem tree, basic commands, file permissions and `chmod` in symbolic and octal form.
- 103–105.** Editors and IDEs; Git, GitHub and GitLab: repositories and their use; Python package managers and environments.
- 106.** The steps of problem solving; high-level versus low-level languages; interpreter versus compiler.
- 107.** Algorithms and workflow diagrams: each symbol and its meaning, the *what* and the *how*, how a loop and a branch are drawn.

UC.2 — Data analysis and visualisation

- 201–202.** The four data structures and which are mutable; NumPy arrays, indexing, slicing, vectorised operations and boolean indexing.
- 203–204.** Figures, subplots, scatter plots, histograms and bin midpoints, labels and legends; your own functions, multiple returns, lambda and recursive functions.
- 205–206.** Number formatting, `open` and its modes, text and CSV files; pandas Series and DataFrames, named indexes, locating rows.
- 207–208.** Pearson and Spearman, the LM method, linear regression and its uncertainties; χ^2 , optimisation, the reduced χ^2 and what its value tells you.
- 209–210.** Global (Newton, Lagrange) versus point-to-point interpolation (linear, spline); NaN handling; what changes outside the range of the data.

UC.3 — Linear algebra, matrices, and images in physics

- 301–302.** Linear systems: the graphical solution, Cramer's rule, $\mathbf{Ax} = \mathbf{b}$, sympy; LU, Gauss elimination, Gauss–Jordan and Gauss–Seidel.
- 303–304.** Non-linear systems with `fsolve()` and `root()`, and why the guess matters; eigenvalues and eigenvectors, the QR method.

305–306. Images as matrices: rasters, colour channels, `imshow()`, pixel histograms; thresholding, tagging the regions, areas and distances in physical units.

307. Scientific data formats: binary, NetCDF, HDF5, VTK and FITS — the structure of each file, its header keys, and how to read them in Python.

Part B — Tutorials and sheets to check

Go over these tutorials. Each tutorial has its worked version beside it as `..._sol.ipynb`.

UC.1:

[tutorials/05-Workflow-diagrams.pdf](#)
[quiz-practice/quiz1_practice_cp1.pdf](#)

UC.2:

[tutorials/01-Input-output-regression.ipynb](#)
[tutorials/02-Time-series.ipynb](#)
[quiz-practice/quiz2_practice.pdf](#) and [homework/homework1.ipynb](#)

UC.3:

[tutorials/03-Linear-and-nonlinear-eqs.ipynb](#)
[tutorials/04-Image-analysis.ipynb](#)

Part C — Pseudocode and workflow diagrams

For each problem, write the pseudocode and, beside it, a workflow diagram with the symbols of Lecture 107. Mark the data I/O, the operations, the decisions and the loops, and name every function with its inputs and its outputs.

1. **A χ^2 fit with confidence intervals.** A file holds N triples (x_i, y_i, σ_i) and a model $f(x; a, b)$. Define $\chi^2(a, b)$, minimise it, report the reduced χ^2 , and map the χ^2 surface around the minimum to draw the contours. Say how the search knows it has finished.
2. **Missing data and interpolation.** A column of measurements has gaps recorded as NaN. Locate the gaps, build an interpolator from the valid points only, fill the gaps, and plot the original and the filled data together so that the filled points can be told apart.
3. **Measuring an object in an image.** A bright object sits on a dark background; one pixel is δ micrometres across. Read one channel, use its pixel histogram to choose a threshold, label the regions of the binary image and keep the largest, and return its area in pixels and in μm^2 .
4. **Reading a scientific data format.** A FITS file holds a 2D image and a header. Open it, return the image and one header key, convert the pixel axes into physical units with that key, and save a labelled two-panel figure in both unit systems.

Important dates: Midterm Exam

- **Midterm Written Exam:** Wednesday 7 October 2026 (at 9am, in class)
- **Notebook Submission Deadline:** Thursday 8 October 2026 (by 9pm)
- **Credits:** 25 points